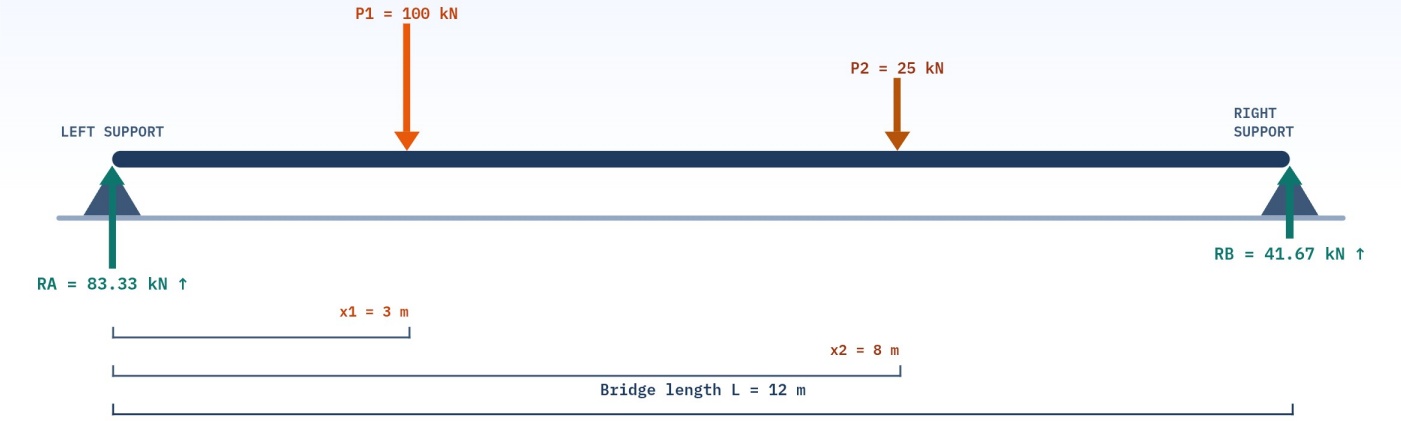
AX = B • Matrix Inverse • Live Analysis

SMART BRIDGE LOAD VISUALIZER

Matrix-Based Support Reaction Analysis



Bridge Parameters

Bridge Span L (m) — 12

Load P1 (kN) — 100

Load P1 position x1 (m) — 3

Load P2 (kN) — 25

Load P2 position x2 (m) — 8

Load Known Test Case | Reset | Try Changing the Load

LEFT SUPPORT
RA
83.33 kN

RIGHT SUPPORT
RB
41.67 kN

Live Load Balance

TOTAL APPLIED LOAD
P1 + P2
125 kN

TOTAL SUPPORT REACTION
RA + RB
125 kN

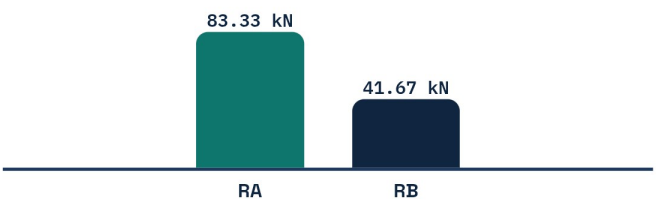
✓ LOAD BALANCE VERIFIED

Validation Check

Total Applied Load = P1 + P2 = 125 kN
Total Support Reaction = RA + RB = 125 kN

✓ VALIDATION PASSED — Total load = total support reactions

Reaction Comparison



Load Position Effect

When a load moves closer to a support, that support generally carries a larger share of the load.

P1 is closer to the left support, so that support carries more of it: left 75 kN, right 25 kN.

P2 is closer to the right support, so that support carries more of it: left 8.33 kN, right 16.67 kN.

Overall, the left support carries more (83.33 kN versus 41.67 kN).

▼ See the Mathematics

Two facts about the bridge: the forces add up (RA + RB = P1 + P2), and the turning effects (moments) about the left support balance (RB × L = P1x1 + P2x2). Written as a matrix equation, **AX = B**.

Step 1 — AX = B

$$\underset{A}{\begin{pmatrix} 1 & 1 \\ 0 & 12 \end{pmatrix}} \underset{X}{\begin{pmatrix} RA \\ RB \end{pmatrix}} = \underset{B}{\begin{pmatrix} 125 \\ 500 \end{pmatrix}}$$

Step 2 — X = A^{-1}B (A^{-1} = (1/L) × [[L, −1],[0, 1]], det A = L = 12)

$$\underset{X}{\begin{pmatrix} RA \\ RB \end{pmatrix}} = \underset{A^{-1}}{\begin{pmatrix} 1 & -0.08 \\ 0 & 0.08 \end{pmatrix}} \underset{B}{\begin{pmatrix} 125 \\ 500 \end{pmatrix}} = \underset{X}{\begin{pmatrix} 83.33 \\ 41.67 \end{pmatrix}}$$

Check by hand: RB = (P1x1 + P2x2) / L = 500 / 12 = 41.67 kN, and RA = P1 + P2 − RB = 125 − 41.67 = 83.33 kN.